Observed Monitoring Metrics — ML-Based Traffic Speed Prediction

Companion document to the paper "Cloud-Fog-Edge Deployment of ML-Based Traffic Speed Prediction: An Empirical Analysis of Performance, Cost, and Scalability". It reports the resource utilisation, thermal, and power metrics observed during the experiments, which the paper references but does not present in full.

Collection methodology

A background monitoring process sampled system metrics every 0.5 seconds during each experiment. CPU variables: system-wide utilisation (%), per-process utilisation (%), operating frequency (MHz), package temperature (deg C), and (where accessible) power (W) via Intel RAPL. GPU variables (via nvidia-smi, on GPU-equipped platforms only): utilisation (%), memory use (MB), temperature (deg C), and power (W).

Reliability caveats (important)

1. **Attribution.** CPU-system and GPU readings are whole-node/whole-device: on shared or virtualised platforms (CSF3 nodes, AWS EC2) they may include activity not attributable to the experiment process. They should be read as qualitative context, not precise per-process measurements; repetition of runs does not remove this systematic attribution error. Only the Dell Precision platform (dedicated hardware, no virtualisation) provides fully representative CPU and GPU readings.
2. **Power.** CPU power via Intel RAPL was accessible only on the Dell Precision (average CPU power 10-46 W across tiers; 0.1-8.7 Wh per training run). On CSF3 the powercap energy counters are root-restricted; AWS EC2 does not expose hardware power to guests. GPU power (nvidia-smi) is available on the GPU platforms.
3. **Diagnostic value.** The telemetry proved essential for outlier attribution: inflated fog-platform runs in an early campaign showed mean CPU clocks of 0.5-1.6 GHz at low package temperature (the signature of an OS power-saving state, not thermal throttling or contention) and were re-measured on mains power. On AWS t2.medium, elevated steal time and run-position-dependent slowdowns identified burstable CPU-credit exhaustion.

Observed metrics by platform and sensor tier

Values aggregate the repeated runs of each platform's initial-training batch: mean of per-run averages; "max" is the maximum across runs. Empty cells: metric not available on that platform.

CSF3 Cloud GPU (A100)

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	19.4	147.2	2,892	45.0/46.8	6.1	16,730	65.2/83.2	25.0/26.0
small	16.7	134.7	2,889	44.7/48.0	7.5	16,834	65.3/67.8	25.6/26.0
medium	16.6	130.1	2,887	43.7/47.2	7.5	16,861	65.0/81.9	24.8/26.0
full	15.7	125.2	2,873	42.6/45.8	7.2	16,907	64.9/68.1	24.5/25.0

Dell Precision Fog GPU (RTX 4090)

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	13.1	180.0	1,967	83.3/94.0	5.4	14,373	28.7/31.9	56.1/58.0
small	10.9	150.2	2,189	86.3/100.0	7.4	14,450	30.3/32.5	59.2/60.0
medium	11.1	152.0	2,163	86.6/100.0	6.9	14,477	30.7/33.4	59.7/62.0
full	10.0	134.1	2,127	86.3/100.0	6.7	14,607	30.4/34.0	59.6/61.0

CSF3 Jetson Sim

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	9.5	127.2	2,785	47.7/49.5	4.8	8,596	68.1/70.4	25.9/26.0
small	9.3	123.5	2,801	49.4/51.0	6.3	8,643	68.7/81.1	26.5/27.0

CSF3 Edge RSU Sim

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	62.4	93.1	2,250	84.2/85.0	—	—	—/—	—/—
small	60.5	90.8	2,250	84.1/86.2	—	—	—/—	—/—

CSF3 RPi Sim

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	85.9	131.5	2,250	84.7/86.0	—	—	—/—	—/—
small	78.0	138.2	2,250	84.3/86.6	—	—	—/—	—/—

AWS t2.medium RSU

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	64.9	128.9	2,300	—/—	—	—	—/—	—/—
small	63.6	125.9	2,300	—/—	—	—	—/—	—/—

Interpretation notes

- CPU process utilisation above 100% indicates multi-threaded data loading and feature engineering phases; training itself is GPU-bound on GPU platforms (GPU utilisation averages are low because per-step kernels for a ~142k-parameter MLP are dominated by launch overhead, consistent with the paper's finding that tensor-core throughput is not the bottleneck).
- On the fog laptop workstation, sustained combined CPU+GPU load settles at reduced CPU clocks (1.2-1.7 GHz) at 80-88 deg C: normal power sharing that does not measurably affect GPU-bound wall-clock times.
- The AWS t2.medium rows reflect a burstable instance: sustained full-CPU work beyond ~1.5 hours exhausts CPU credits and caps the instance at its baseline rate; short workloads run at burst speed. Deployment schedules on burstable instances should be budgeted at baseline speed.